Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 15-07-2026
Team ID: **23BFA05428**
Project Name: **Smart Library Request Workflow in ServiceNow**
Maximum Marks: 4 Marks

Technical Architecture

The Smart Library Request Workflow is implemented on the ServiceNow cloud platform. Students access the Service Portal to browse the catalog and submit borrowing requests. Catalog Client Scripts validate user input, while GlideAjax and Script Includes verify book availability from custom library tables. Flow Designer automates the approval workflow by routing requests to librarians, updating inventory, recording issue/return transactions, and sending notifications. Administrators monitor reports and dashboards for borrowing statistics, overdue books, and inventory utilization. The architecture is modular, secure, and scalable, allowing institutions to manage large collections and users efficiently.

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Component	Purpose	Technology
Service Portal	Book search & requests	Service Portal
Service Catalog	Request creation	Catalog Items
Client Scripts	Form validation	JavaScript
Server Logic	Availability validation	GlideAjax, Script Includes
Workflow	Approvals	Flow Designer
Database	Books, Members, Requests	Custom Tables
Notifications	Email alerts	Notification Engine
Analytics	Reports	Dashboards

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Technology Stack - Application Characteristics

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The application follows ServiceNow best practices by using reusable workflows, role-based access control, cloud-hosted infrastructure, and centralized data storage. Security is enforced through ACLs, authenticated sessions, and HTTPS communication. The solution is designed to scale with increasing numbers of books, members, and requests while maintaining high performance and availability.

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Characteristic	Description
Security	Role-based ACLs, user roles, HTTPS
Scalability	Supports thousands of users and books
Availability	24×7 cloud deployment
Performance	Optimized workflows and server-side processing
Maintainability	Reusable business rules and script includes